Homework 12

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13.1.2

For a given n, $l = 0, \cdots, n-1$;

for a given l, $m = -l, -l+1, \cdots, l$ $\Rightarrow$ $(2l+1)$ terms.

Therefore, the degeneracy is

$$\sum_{l=0}^{n-1}(2l+1) = 2\sum_{l=0}^{n-1} l + \sum_{l=0}^{n-1} 1$$

$$= 2 \cdot \frac{(0+n-1)n}{2} + n$$

$$= n^2 .$$

13.1.3

$$\psi_{210}(r,\theta,\varphi) = R_{21}(r)\, Y_1^0(\theta,\varphi) = \frac{U_{21}(r)}{r} Y_1^0(\theta,\varphi)$$

$U_{21}(r) = e^{-\rho} v_{21}$, where $\rho = \sqrt{-2mE/\hbar^2}\, r \equiv \frac{1}{na_0} r$

$v_{El} = \rho^{l+1} \sum_{k=0}^{\infty} C_k \rho^k$. By recursion relation, k should terminate at $n-l-1$.

In this problem, $n=2$, $l=1$, we have

$v_{21} = \rho^2 C_0 \rho^0 = C_0 \rho^2$. (only $k=0$ term left)

Therefore,

$$\psi_{210} = \frac{1}{\rho} e^{-\rho} (\rho^2 C_0)\, Y_1^0(\theta,\varphi) = C_0 \rho e^{-\rho} Y_1^0(\theta,\varphi), \quad \rho = \frac{1}{2a_0} r$$

Normalization:

$$\int \psi_{210}^* \psi_{210}\, r^2 dr\, d\Omega = 1$$

$$\Rightarrow (2a_0)^3 C_0^2 \int_0^\infty \rho^2 e^{-2\rho} \rho^2 d\rho \underbrace{\int Y_1^{0*}(\theta,\varphi) Y_1^0(\theta,\varphi)\, d\Omega}_{=1} = 1$$

$$\Rightarrow 1 = (2a_0)^3 C_0^2 \int_0^\infty \rho^4 e^{-2\rho} d\rho$$

$$= (2a_0)^3 C_0^2 \int_0^\infty (2\rho)^4 e^{-2\rho} d(2\rho) \,/\, 32$$

$$= \frac{1}{32}(2a_0)^3 C_0^2 \cdot 4!$$

$$= 6 a_0^3 C_0^2$$

$$\Rightarrow C_0 = \left(\frac{1}{6a_0^3}\right)^{\frac{1}{2}}$$

Thus,

$$\psi_{210}(r,\theta,\varphi) = \left(\frac{1}{6a_0^3}\right)^{\frac{1}{2}} \left(\frac{r}{2a_0}\right) e^{-r/2a_0}\, Y_1^0(\theta,\varphi) \qquad * \; Y_1^0(\theta,\varphi) = \left(\frac{3}{4\pi}\right)^{\frac{1}{2}} \cos\theta$$

$$= \left(\frac{1}{32\pi a_0^3}\right)^{\frac{1}{2}} \frac{r}{a_0} e^{-r/2a_0} \cos\theta .$$

13.1.5

According to Ehrenfest's theorem,

$$\langle \dot{\Omega} \rangle = \frac{1}{i\hbar} \langle [\Omega, H] \rangle = 0$$

$$\Rightarrow \quad 0 = \langle [\Omega, H] \rangle$$

$$= \langle [\vec{R}\cdot\vec{P}, H] \rangle$$

$$= \langle [\vec{R}\cdot\vec{P}, \frac{\vec{P}^2}{2m} - \frac{e^2}{r}] \rangle$$

$$= \langle [\vec{R}, \frac{\vec{P}^2}{2m}]\cdot\vec{P} + \vec{R}\cdot[\vec{P}, -\frac{e^2}{r}] \rangle \qquad (1)$$

$*$ $[x, p] = i\hbar$

$[x, f(p)] = i\hbar \frac{df}{dp}$

$[g(x), p] = i\hbar \frac{dg}{dx}$.

$$[\vec{R}, \frac{\vec{P}^2}{2m}] = i\hbar \cdot \frac{\vec{P}}{m}$$

$$[\vec{P}, -\frac{e^2}{r}] = [p_x\hat{i} + p_y\hat{j} + p_z\hat{k}, -\frac{e^2}{\sqrt{x^2+y^2+z^2}}]$$

$$[p_x, -\frac{e^2}{\sqrt{x^2+y^2+z^2}}] = i\hbar \frac{d}{dx}\left(\frac{e^2}{\sqrt{x^2+y^2+z^2}}\right)$$

$$= i\hbar \frac{xe^2}{(x^2+y^2+z^2)^{\frac{3}{2}}}$$

$$= i\hbar \frac{xe^2}{r^3}$$

Similarly, we have

$$[p_y, -\frac{e^2}{\sqrt{x^2+y^2+z^2}}] = i\hbar \frac{ye^2}{r^3}$$

$$[p_z, -\frac{e^2}{\sqrt{x^2+y^2+z^2}}] = i\hbar \frac{ze^2}{r^3}$$

Therefore,

$$[\vec{P}, -\frac{e^2}{r}] = i\hbar \frac{e^2}{r^3}(x\hat{i} + y\hat{j} + z\hat{k})$$

$$= i\hbar e^2 \frac{\vec{R}}{r^3}$$

Now, (1) can be rewritten as

$$\langle i\hbar \frac{\vec{P}}{m}\cdot\vec{P} + \vec{R}\cdot i\hbar e^2 \frac{\vec{R}}{r^3} \rangle = 0$$

$$\Rightarrow \quad \langle \frac{\vec{P}^2}{m} \rangle + \langle \frac{e^2 \vec{R}^2}{r^3} \rangle = 0$$

$$\Rightarrow \quad 2\langle \frac{\vec{P}^2}{2m} \rangle + \langle \frac{e^2}{r} \rangle = 0$$

$$\Rightarrow \quad 2\langle T \rangle + \langle V \rangle = 0$$

$$\Rightarrow \quad \langle T \rangle = -\frac{1}{2}\langle V \rangle.$$

12.5.1

First, apply the rotation to the coordinate.

According to (12.2.9), $\langle xy | I - i \frac{\varepsilon_z L_z}{\hbar} | \psi \rangle = \psi(x + y\varepsilon_z, y - x\varepsilon_z)$.

Here, we have

$$\langle xy | U_L(\varepsilon_z) | \psi \rangle = \begin{pmatrix} \psi_x(x + y\varepsilon_z, y - \varepsilon_z x) \\ \psi_y(x + y\varepsilon_z, y - \varepsilon_z x) \end{pmatrix}$$

$$\langle xy | U_L(\varepsilon_z) | \psi \rangle = \begin{pmatrix} (I - \frac{i\varepsilon_z}{\hbar} L_z)\psi_x(x,y) \\ (I - \frac{i\varepsilon_z}{\hbar} L_z)\psi_y(x,y) \end{pmatrix}$$

$$= \begin{pmatrix} I - \frac{i\varepsilon_z}{\hbar} L_z & 0 \\ 0 & I - \frac{i\varepsilon_z}{\hbar} L_z \end{pmatrix} \begin{pmatrix} \psi_x(x,y) \\ \psi_y(x,y) \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \psi_x(x + y\varepsilon_z, y - \varepsilon_z x) \\ \psi_y(x + y\varepsilon_z, y - \varepsilon_z x) \end{pmatrix} = \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{i\varepsilon_z}{\hbar} \begin{pmatrix} L_z & 0 \\ 0 & L_z \end{pmatrix} \right] \begin{pmatrix} \psi_x(x,y) \\ \psi_y(x y) \end{pmatrix}$$

Now, rotate this vector by the same ε_z.

$$U_S(\varepsilon_z) \begin{pmatrix} \psi_x(x + y\varepsilon_z, y - x\varepsilon_z) \\ \psi_y(x + y\varepsilon_z, y - x\varepsilon_z) \end{pmatrix}$$

$$= \begin{pmatrix} \psi_x(x + y\varepsilon_z, y - x\varepsilon_z) - \psi_y(x + y\varepsilon_z, y - x\varepsilon_z)\varepsilon_z \\ \psi_x(x + y\varepsilon_z, y - x\varepsilon_z)\varepsilon_z + \psi_y(x + y\varepsilon_z, y - x\varepsilon_z) \end{pmatrix}$$

$$= \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{i\varepsilon_z}{\hbar} \begin{pmatrix} 0 & -i\hbar \\ i\hbar & 0 \end{pmatrix} \right] \begin{pmatrix} \psi_x(x + y\varepsilon_z, y - x\varepsilon_z) \\ \psi_y(x + y\varepsilon_z, y - x\varepsilon_z) \end{pmatrix}$$

Multiply these:

$$U_{tot}(\varepsilon) = U_S(\varepsilon_z) U_L(\varepsilon_z)$$

$$= \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{i\varepsilon_z}{\hbar} \begin{pmatrix} 0 & -i\hbar \\ i\hbar & 0 \end{pmatrix} \right] \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{i\varepsilon_z}{\hbar} \begin{pmatrix} L_z & 0 \\ 0 & L_z \end{pmatrix} \right]$$

$$= \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{i\varepsilon_z}{\hbar} \begin{pmatrix} L_z & 0 \\ 0 & L_z \end{pmatrix} - \frac{i\varepsilon_z}{\hbar} \begin{pmatrix} 0 & -i\hbar \\ i\hbar & 0 \end{pmatrix} \right]$$

12.5.3

(1) $J_+ = J_x + i J_y$

$J_- = J_x - i J_y$

$J_x = \frac{1}{2}(J_+ + J_-)$ and $J_y = \frac{1}{2i}(J_+ - J_-)$

$\langle J_\pm \rangle = \langle j m | J_\pm | j m \rangle = 0$

Therefore, $\langle J_x \rangle = \langle J_y \rangle = 0$

(2) By symmetry, we have $\langle J_x^2 \rangle = \langle J_y^2 \rangle$

Since $J^2 = J_x^2 + J_y^2 + J_z^2$, we have

$\langle J^2 \rangle - \langle J_z^2 \rangle = \langle J_x^2 \rangle + \langle J_y^2 \rangle = 2 \langle J_x^2 \rangle$

$\langle J^2 \rangle = j(j+1)\hbar^2$

$\langle J_z^2 \rangle = m^2 \hbar^2$

Therefore,

$\langle J_x^2 \rangle = \langle J_y^2 \rangle = \frac{1}{2}\hbar^2 [j(j+1) - m^2]$

(3) Eqn. 9.2.9 $(\Delta\Omega)^2(\Delta\Lambda)^2 \geq |\langle \psi | \hat{\Omega}\hat{\Lambda} | \psi \rangle|^2$

$$\Delta J_x \cdot \Delta J_y = \sqrt{\langle J_x^2 \rangle - \cancel{\langle J_x \rangle^2}} \cdot \sqrt{\langle J_y^2 \rangle - \cancel{\langle J_y \rangle^2}}$$
$$= \sqrt{\langle J_x^2 \rangle \langle J_y^2 \rangle}$$
$$= \langle J_x^2 \rangle$$
$$= \frac{1}{2}\hbar^2 [j(j+1) - m^2]$$

$$|\langle J_x J_y \rangle| = |\langle \frac{1}{4i} (J_+^2 - J_-^2 - J_+J_- + J_-J_+ \rangle|$$
$$= \frac{1}{4} | \cancel{\langle J_+^2 \rangle} - \cancel{\langle J_-^2 \rangle} + \langle J_-J_+ \rangle - \langle J_+J_- \rangle |$$

(P327)
$$= \frac{1}{4} | \langle J^2 - J_z^2 - \hbar J_z \rangle - \langle J^2 - J_z^2 + \hbar J_z \rangle |$$
$$= \frac{1}{4} | -2\hbar \langle J_z \rangle |$$
$$= \frac{1}{4} | 2\hbar \cdot m\hbar |$$
$$= \pm \frac{1}{2} m\hbar^2 \quad (\text{"+" for } m \geq 0, \text{ "−" for } m < 0)$$

$$\Delta J_x \cdot \Delta J_y - |\langle J_x J_y \rangle| = \frac{1}{2}\hbar^2 [j(j+1) - m(m\pm1)] \geq 0$$

since $-j \leq m \leq j$

$\Rightarrow$ Inequality is satisfied.

(4) If $m = \pm j$, then $j(j+1) - m(m\pm1) = 0$, and the uncertainty bound is saturated.

14.3.3

$$Tr(\sigma_i\sigma_j) = Tr(\sigma_j\sigma_i) \quad \text{property of trace}$$

For $i \neq j$, $\sigma_i\sigma_j = -\sigma_j\sigma_i$ (14.3.32).

Therefore, $Tr(\sigma_i\sigma_j) = -Tr(\sigma_j\sigma_i)$.

$\Rightarrow$ $Tr(\sigma_j\sigma_i) = -Tr(\sigma_j\sigma_i)$

$\Rightarrow$ $Tr(\sigma_j\sigma_i) = 0 = Tr(\sigma_i\sigma_j)$

Meanwhile, $\sigma_i\sigma_j = i\sigma_k$ (14.3.33)

$\Rightarrow$ $Tr(i\sigma_k) = 0$

$\Rightarrow$ $Tr(\sigma_k) = 0$. for all k.

14.3.4

(1)
$$\begin{aligned}
\sigma_i\sigma_j &= \frac{1}{2}\left([\sigma_i,\sigma_j] + [\sigma_i,\sigma_j]_+\right) \\
&= \frac{1}{2}\left(2i\varepsilon_{ijk}\sigma_k + 2\delta_{ij}I\right) \\
&= i\varepsilon_{ijk}\sigma_k + \delta_{ij}I
\end{aligned}$$

$$\begin{aligned}
(\vec{A}\cdot\vec{\sigma})(\vec{B}\cdot\vec{\sigma}) &= \left(\sum_i A_i\sigma_i\right)\left(\sum_j B_j\sigma_j\right) \\
&= \sum_i\sum_j A_iB_j\sigma_i\sigma_j \\
&= \sum_i\sum_j A_iB_j\, i\varepsilon_{ijk}\sigma_k + \sum_i\sum_j A_iB_j\delta_{ij}I \\
&= i(\vec{A}\times\vec{B})\cdot\vec{\sigma} + (\vec{A}\cdot\vec{B})I
\end{aligned}$$

(2) Set $M = (\vec{A}\cdot\vec{\sigma})(\vec{B}\cdot\vec{\sigma}) = \sum_\alpha m_\alpha\sigma_\alpha$

$$\begin{aligned}
m_\beta &= \frac{1}{2}Tr(M\sigma_\beta) \qquad \beta = 0,1,2,3 \qquad \beta=0 \rightarrow \text{Identity} \\
&= \frac{1}{2}Tr[(\vec{A}\cdot\vec{\sigma})(\vec{B}\cdot\vec{\sigma})\sigma_\beta] \\
&= \frac{1}{2}Tr\left(\sum_i\sum_j A_iB_j\sigma_i\sigma_j\sigma_\beta\right)
\end{aligned}$$

For $\beta = 0$, $\sigma_\beta = I$

$$\begin{aligned}
m_0 &= \frac{1}{2}Tr\left(\sum_i\sum_j A_iB_j\sigma_i\sigma_j\right) \\
&= \frac{1}{2}\sum_i\sum_j A_iB_j\,Tr(\sigma_i\sigma_j) \qquad * \; Tr(\sigma_i\sigma_j) = 2\delta_{ij} \\
&= \sum_i\sum_j A_iB_j\delta_{ij} \\
&= \vec{A}\cdot\vec{B}
\end{aligned}$$

For $\beta \neq 0$

$$m_\beta = \frac{1}{2} Tr\left(\sum_i\sum_j A_i B_j \sigma_i \sigma_j \sigma_\beta\right)$$

$$= \frac{1}{2} \sum_i\sum_j A_i B_j \, Tr(\sigma_i \sigma_j \sigma_\beta) \qquad * \; \sigma_i\sigma_j = i\,\varepsilon_{ijk}\sigma_k$$

$$= \frac{1}{2} \sum_i\sum_j A_i B_j \, Tr(i\,\varepsilon_{ijk}\sigma_k\sigma_\beta)$$

$$= \frac{1}{2} \sum_i\sum_j A_i B_j \cdot i\,\varepsilon_{ijk} \, Tr(\sigma_k\sigma_\beta) \qquad * \; Tr(\sigma_k\sigma_\beta) = 2\delta_{k\beta}$$

$$= \sum_i\sum_j A_i B_j \cdot i\,\varepsilon_{ijk}\,\delta_{k\beta}$$

$$= i\,(\vec{A}\times\vec{B})_\beta$$

$$\Rightarrow \sum_{\beta\neq 0} m_\beta\sigma_\beta = i\,(\vec{A}\times\vec{B})\cdot\vec{\sigma}$$

$$\Rightarrow M = \sum_\beta m_\beta\sigma_\beta = m_0 I + i(\vec{A}\times\vec{B})\cdot\vec{\sigma}$$

$$= (\vec{A}\cdot\vec{B})I + i\,(\vec{A}\times\vec{B})\cdot\vec{\sigma}$$